An Adjoint Gradient Reaches a Worst-of-Band $|S_{11}|$ of $-9.5\,\mathrm{dB}$ with an Order of Magnitude Fewer Solves than Gradient-Free Search: Copper-Only Design of a 2.4 GHz Conformal Patch Antenna

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Abstract

We design the copper of a probe-fed 2.4 GHz patch antenna on a standard 1.6 mm FR-4 board bent to a 100 mm radius, with 38 shape parameters and a finite-element solver, geode-fem, whose discrete shape adjoint gives the gradient of the band objective at one factorization per frequency. On the coarse optimization model the adjoint reaches a worst-of-band $|S_{11}|$ of $-9.5\,\mathrm{dB}$ in 35 solves; separable CMA-ES and NEWUOA driven by the independent solver Palace need 11–23 times as many, and an estimated 7.6 to 15.5 times the wall-clock on one host. Palace's forward solve is about 11% faster than geode-fem's, so the advantage comes from the gradient. A four-parameter rectangle, given the final frequency correction in advance, also meets the target on the second-order model. The forward model's error sits at the patch rim. After one more frequency space-mapping step, the design with a 1.6 mm probe reaches $-10.46\,\mathrm{dB}$ worst in band on Palace with third-order elements, with a 93.0 MHz $-10\,\mathrm{dB}$ band covering the ISM band by 6.5 and 2.9 MHz; geode-fem's finest mesh, not used in choosing the design, agrees. The lower margin covers the order-1 estimate of the remaining drift (6.4 MHz), which may be low, but not a rim-size fit (13.5 MHz), at which 2.400 GHz would sit at $-8.94\,\mathrm{dB}$.

1 Introduction

Gradient-based inverse design is routine in nanophotonics, where differentiable solvers such as CEVICHE (Hughes et al., 2019), the Meep adjoint solver (Hammond et al., 2022), SPINS-B (Su et al., 2019) and FDTDX (Mahlau et al., 2026) optimize a permittivity density on a structured grid. Shape derivatives of scattering parameters for finite-element models with ports are older (Nikolova et al., 2004a; Webb, 2002; Webb and Akel, 2000) and are available in commercial tools (Ansys, Inc., 2025; COMSOL AB, 2023). A gradient is worth its implementation cost only if it buys something a gradient-free method does not. For a design with two nearly independent parameters, finite differences or a scaling law are cheap and a gradient adds little. The case for a gradient is a design with many coupled parameters, where finite differences cost one solve per parameter and gradient-free search scales poorly with dimension.

This paper measures that case on one antenna. A 2.4 GHz probe-fed patch is bent around a cylinder of radius $R_b = 100\,\mathrm{mm}$. The board is a uniform 1.6 mm FR-4 shell that is not designed; 38 smooth parameters move the copper outline, the corners and the feed. geode-fem, a differentiable finite-element solver with a discrete shape adjoint, optimizes the band objective by L-BFGS. We run the same problem, objective, parameters and volume guard with two gradient-free optimizers, separable CMA-ES (Hansen and Ostermeier, 2001; Ros and Hansen, 2008) and NEWUOA (Powell, 2006; Zhang et al., 2023), each with geode-fem and with Palace (Palace developers, AWS Center for

Quantum Computing, 2023) as the forward solver, and count solves and wall-clock to fixed objective levels on one host.

Three results follow.

- **The gradient makes the search about an order of magnitude cheaper** (Section 5). To reach −9.5 dB on the optimization model the adjoint uses 35 solves; the gradient-free runs use 385 to 805 (11–23×) and an estimated 7.6 to 15.5 times the wall-clock (geode-fem's side is its solves times its measured per-solve cost on the same host). Palace's forward evaluation is cheaper than geode-fem's, so the difference is the gradient, not the solver. Given enough solves, one CMA-ES seed of three finds a design as good as the adjoint's, and a four-parameter rectangle, given the final band shift in advance, also meets the target on the second-order model, so the gradient buys speed rather than a better optimum, and the 38-parameter basis is not what makes the target reachable.
- **The forward model's error sits at the patch rim** (Section 4). Refining the rim moves the resonance; refining the substrate interior does not. Palace on geode-fem's own meshes reproduces geode-fem at first and second order, and Palace at third order agrees with geode-fem's finest rim-refined mesh to under 1 MHz.
- **A design re-centred on the refined model meets the ISM band at −10 dB on both fine models** (Section 7), with a probe of realizable width (1.6 mm, modelled as a lumped port) and margins small enough that we state which drift estimates they cover.

We also report the checks around the design. The bend moves the resonance by under 0.5 MHz, though it changes the match by about 1 dB; the absorber and air margin move the band by at most a few megahertz (Section 4); and the radiation efficiency of a closely related design is 0.62 with perfectly conducting copper (Section 7).

Evidence tiers. We keep three tiers apart. A *bounded* result is a proven two-sided bound; no result in this paper is of that kind. A *model-dependent* result is a value computed for a stated discrete model (mesh, element order, absorber and port model). An *estimated* result is an extrapolation to the mesh limit, given with its band and never called a converged value.

2 Related Work

Finite-element S-parameter sensitivities with ports. Design sensitivities of the scattering matrix for tetrahedral edge elements were derived by Webb and Akel (2000), extended to the frequency response by Webb (2002) and used in gradient optimization of microwave devices by Nair and Webb (2003). Adjoint and self-adjoint formulations are reviewed by Nikolova et al. (2004a) and Nikolova et al. (2006b), and continuum shape sensitivity for high-frequency devices is given by Choi et al. (2011). HFSS computes S-parameter derivatives with respect to geometric parameters by reusing the LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL documents adjoint shape optimization of waveguide devices on a deformed mesh (COMSOL AB, 2023; Jensen, 2022). For antennas, Ghassemi et al. (2013) evolve a microstrip antenna with an FEM adjoint sensitivity and a small set of control vertices.

Variable-fidelity antenna optimization and space mapping. Space mapping aligns a cheap coarse model with an accurate fine model and optimizes the coarse one (Bandler et al., 1994, 2004; Koziel et al., 2006). In antenna design the coarse model is often the same electromagnetic solver on a coarse discretization, corrected by frequency scaling or by response-shape correction (Koziel,

2010; Koziel and Ogurtsov, 2014), and adjoint sensitivities have been used inside trust-region space mapping (Koziel et al., 2013). Our correction is the simplest member of this family: a frequency shift fitted between two discretizations of the same solver.

Automated shape differentiation. Shape derivatives of finite-element models can be generated automatically: in UFL and Firedrake (Dokken et al., 2020; Ham et al., 2019; Paganini and Wechsung, 2021) and in NGSolve (Gangl et al., 2021). Wang and Anderson (2011) give a discrete Maxwell shape adjoint on unstructured meshes in two dimensions, in the time domain and without ports. geode-fem obtains the element Jacobians by forward-mode differentiation of its closed-form element kernels inside a tensor stack (Tracel AI, 2024).

Structured-grid design of metallic antennas. Density-based topology optimization of conductors on a fixed grid goes back at least to Erentok and Sigmund (2011); Hassan et al. (2014b) and Hassan et al. (2014a) optimize the patch and ground of coax-fed microstrip antennas with an FDTD adjoint, and Nikolova et al. (2004b, 2006a) give FDTD shape sensitivities. A plain Yee grid staircases a curved conductor (Cangellaris and Wright, 1991; Häggblad and Runborg, 2014), but locally conformal FDTD handles curved PEC (Dey and Mittra, 1997; Mittra and Yu, 2000), including uniformly stable second-order schemes (Benkler et al., 2006; Zagorodnov et al., 2003). Subpixel-smoothed projection gives the Meep density adjoint subpixel-accurate boundaries (Hammond et al., 2025; Romano et al., 2026), though the material-grid adjoint excludes metals (Meep developers, 2026); Tidy3D offers autograd shape gradients and conformal PEC averaging (Flexcompute, Inc., 2026); and Hooten et al. (2025) and Liu and Bonar (2024) map shape parameters to a fixed grid. Boundary-integral methods carry recent PEC shape-sensitivity work (Arens et al., 2026; Balouchev et al., 2024; Takahashi, 2025).

Conformal and cylindrical patch antennas. The cylindrical-rectangular patch has a cavity-model analysis (Luk et al., 1989), and nonplanar microstrip antennas and conformal arrays are treated in the monographs of Wong (1999) and Josefsson and Persson (2006). We use a full-wave model and do not compare with these analyses; the curvature here is mild (Section 4).

Gradient-free optimizers. CMA-ES adapts a full covariance matrix (Hansen and Ostermeier, 2001); its separable variant restricts it to the diagonal, which makes each update linear in the dimension (Ros and Hansen, 2008). NEWUOA builds a quadratic interpolation model inside a trust region (Powell, 2006); we use the PRIMA implementation (Zhang et al., 2023).

Position of this work. The closest prior work is the FEM S-parameter sensitivity line and the variable-fidelity antenna literature, which already reuse factorizations, differentiate with respect to geometry and correct coarse models. What this paper adds is a measurement: on a 38-parameter open-radiator problem with a moving feed, the solve and wall-clock cost of the adjoint against two gradient-free optimizers driven by two independent solvers, and a design checked on two fine models, one of them from an independent solver (which informed the final band shift). We do not claim a first EM shape adjoint, and we make no performance comparison with structured-grid methods.

3 Method

3.1 The open-radiator forward model

The forward problem is the driven frequency-domain curl-curl equation, discretized with Nédélec edge elements of order $p = 1$ or $p = 2$ on an unstructured tetrahedral mesh. Lengths are in

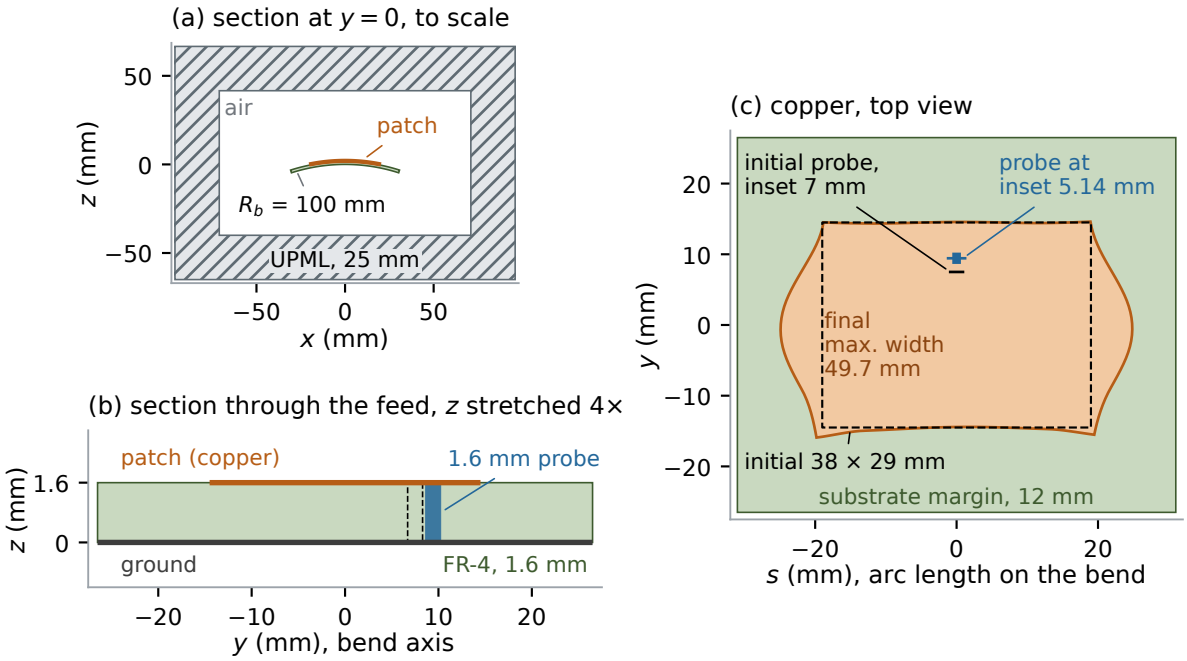


Figure 1: The fixture and the final copper design. (a) Section at $y = 0$, to scale: the FR-4 board (green) on the $R_b = 100\,\text{mm}$ cylinder with the patch on top, inside the air box and the 25 mm box UPML (hatched). (b) Section through the feed at $s = 0$, vertical axis stretched four times: patch, 1.6 mm FR-4, ground, and the 1.6 mm probe, final (filled) and initial (dashed). (c) Top view in the chart (s, y): the initial $38 \times 29\,\text{mm}$ patch (dashed), the final copper (filled) and the probe positions (bars). The board is the same in both. Drawn from `baseline_2g4_L1.toml` and `recentre/geometry/rbG_L1_copper_38_sm57_w16.json`.

millimetres and the solver works with $\omega = k_0 = 2\pi f/c$ in rad/mm. The assembled system is the complex-symmetric pencil

$$A(\omega)\,\mathbf{x} = \left[\, K(\nu) - \omega^2 M(\varepsilon) + \tfrac{j\omega}{Z_s}\, S_p \,\right] \mathbf{x} = \mathbf{b}, \tag{1}$$

with curl-curl stiffness K, mass matrix M and lumped-port surface term S_p. The radiating boundary is a box UPML of thickness $t = 25\,\text{mm}$: per-element tensor weights Λ enter K (as $\nu = \Lambda^{-1}$) and M (as $\varepsilon = \varepsilon_r \Lambda$), with the stretch $\varsigma = 1 - j\sigma_0 (d/t)^2/\omega$ per axis at depth d and $\sigma_0 = 1$. The FR-4 has $\varepsilon_r = 4.4$ and $\tan\delta = 0.02$, and the conductors are perfect electric conductors (PEC). The port is a rectangle of width w_p along y and length $l_p = 1.6\,\text{mm}$ across the substrate, with sheet impedance $Z_s = R_0\, w_p/l_p$ and $R_0 = 50\,\Omega$. The reflection coefficient follows from the port voltage V:

$$S_{11} = \frac{Z - R_0}{Z + R_0}, \qquad Z = \frac{V}{I}, \qquad I = \frac{2V_{\text{inc}} - V}{R_0}. \tag{2}$$

Every evaluation checks passivity, $|S_{11}| \le 1$.

3.2 Shape adjoint with one factorization per frequency

The design moves mesh node coordinates X, so the gradient needs $\partial K/\partial X$, $\partial M/\partial X$, $\partial S_p/\partial X$ and $\partial \mathbf{b}/\partial X$. geode-fem evaluates its closed-form element kernels in first-order dual arithmetic: seeding the tangent of one node coordinate returns the exact partial derivatives of every local matrix entry.

Table 1: The 38 copper parameters, per unit coefficient. The board thickness has none.

Group	Count	What moves
Outline	32	8 clamped cubic B-splines per patch edge ($\pm y$, $\pm s$); each moves its edge outward in the surface by 1 mm, ramped in from the feed band and decaying over the 12 mm substrate margin
Corners	4	one bump per corner, moving it outward along the bisector by 1 mm (negative values chamfer), support $1 - \mathrm{smoothstep}(d/6\,\mathrm{mm})$
Feed	2	the probe's position along y (its inset from the $+y$ edge) and its width w_p, each 1 mm

The UPML weights are held fixed under node motion. Because $A(\omega)$ is complex symmetric, the adjoint system uses the same operator, and one LU factorization per frequency serves the forward and the adjoint solve. The nodal gradient is $\lambda^{\top}(\partial A/\partial X)\,\mathbf{x}$ plus the explicit dependence of the port drive and of the port-voltage functional on the geometry. When a parameter moves the feed, the port faces move and the port is rebuilt on every moved mesh. A unit test of the moving-feed gradient on a curved mesh, with a source that varies across the port so that the explicit term matters, matches central finite differences to below 10^{-8}; without the explicit term it misses by a relative error of about 3.

3.3 Copper design freedom on a fixed curved board

The board is not a design variable. Each substrate node has chart coordinates (s, y), with $s = R_b\,\mathrm{atan2}(x, z - z_{\mathrm{axis}})$ the mid-plane arc length and y the bend axis. A design applies a chart displacement $(\Delta s, \Delta y)$ and moves the node to angle $\phi_0 + \Delta s/R_b$ and height $y_0 + \Delta y$ at its own radius, so the FR-4 stays a uniform 1.6 mm shell. Every run records the largest change of any substrate node's radius ($\leq 1.4 \times 10^{-14}$ mm). Air nodes follow a harmonic extension of the motion; the UPML and the outer boundary are fixed. The design gradient chains the nodal adjoint gradient through the exact Jacobian of this map.

Table 1 lists the 38 parameters. They are smooth functions of the chart, so the same coefficient vector defines the same copper on every mesh of the family, which lets a design be carried between discretizations. The basis cannot open a hole in the patch; slots change the conductor's topology and are outside it.

3.4 Objective, optimizer and guards

The objective is a soft maximum of $|S_{11}|^2$ over $n_f = 5$ band frequencies,

$$G(\mathbf{c}) = \frac{1}{\beta} \ln \sum_f \exp\bigl(\beta\,|S_{11}(f;\mathbf{c})|^2\bigr), \qquad \beta = 25, \tag{3}$$

over the coefficient vector $\mathbf{c}$. G exceeds the largest $|S_{11}|^2$ by at most $\ln 5/\beta = 0.064$, and every level reported in this paper is the worst of the band points, not G. L-BFGS (memory 8) with Armijo backtracking and a cap of 1 coefficient unit per step minimizes G. A per-element guard rejects any trial step that would shrink a tetrahedron's signed volume below 0.25 of its base value. Every descent reported here ends on the guard with the gradient still nonzero (one intermediate re-based run of the space-mapping chain ended on a plateau), so the designs are guard-limited rather than stationary.

Before and after each run the pipeline gradient is compared with a central finite difference along the steepest-descent direction and along random directions. The worst relative error over the runs

Table 2: Where the error sits: the initial patch's $|S_{11}|$ dip under refinement of L1, geode-fem at $p = 1$. "Slab" is the element size in the substrate under the patch, "rim" the mean edge length within 0.5 mm of the patch edge, "layers" the median number of elements through the substrate under the patch. Model-dependent values.

Mesh	Slab (mm)	Rim (mm)	Layers	DOFs	Dip (GHz)	vs L1 (MHz)
L1	0.80	0.78	2.00	150,561	2.3118	0
slab 0.533	0.533	0.52	2.54	234,691	2.3273	+15.5
slab 0.40	0.40	0.39	3.44	414,837	2.3404	+28.6
slab 0.30	0.30	0.29	4.19	828,775	2.3480	+36.2
rim 0.40	0.80	0.39	1.99	317,424	2.3420	+30.2
rim 0.20	0.80	0.20	1.99	430,808	2.3569	+45.1
slab 0.30 + rim 0.10	0.30	0.10	4.19	1,738,557	2.3659	+54.1

reported here is 1.5×10^{-5} (tolerance 5×10^{-3}), and a per-group check on the two coarsest meshes passes to 1.9×10^{-7}. The adjoint is implemented for $p = 1$ only, so optimization runs at $p = 1$ and finer models enter through verification.

4 Fixture and Forward-Model Accuracy

Antenna. The initial antenna is a 38×29 mm PEC patch (width along s, resonant length along y) on 1.6 mm FR-4 with a 12 mm substrate and ground margin, fed by a 1.6 mm probe, modelled as a uniform lumped port, inset 7.0 mm from the $+y$ edge (Figure 1). The bend axis is parallel to y, so the curvature runs along the non-radiating width; the sagitta at the patch edge is 1.8 mm. A 40 mm air margin separates the board from the UPML. The target is $|S_{11}| \leq -10$ dB over the 2.400–2.4835 GHz ISM band.

The coarse mesh. The optimization mesh L1 has 150,561 edges, a median of 2.0 element layers through the substrate under the patch and a mean edge length of 0.78 mm at the patch rim. On L1 at $p = 1$ the initial patch's $|S_{11}|$ dip is at 2.3118 GHz; at $p = 2$ on L1 (P2/L1) it is at 2.3566 GHz, 45 MHz higher. A direct LU solve at P2/L1 (804,460 unknowns) takes 132 GiB in geode-fem, so refining further at $p = 2$ is not practical with this solver.

The error sits at the rim. Table 2 refines L1 in two ways. Refining the whole slab under the patch to 0.40 mm gives 3.4 layers and, as a side effect, a 0.39 mm rim. Refining only within 0.5 mm of the patch edge to the same rim size, with the 2-layer interior unchanged, lands within 1.6 MHz of it (2.3404 against 2.3420 GHz). The interior layer count does not decide the dip; the rim does, as expected where the field is singular at a conductor edge (Meixner, 1972). A 0.2 mm rim at $p = 1$ (431k unknowns) matches P2/L1 (805k) to 0.3 MHz. Over all seven $p = 1$ rungs the dip follows $f = f_\infty - C\, h_{\mathrm{rim}}^{0.80}$ with a 1.3 MHz rms residual (Figure 2b). The exponent is a fit to the rim edge length on graded meshes, not a demonstrated asymptotic order. The combined rung, which halves the rim to 0.1 mm and refines the slab, moves the dip 9.0 MHz.

An independent solver on the same meshes. Palace (Palace developers, AWS Center for Quantum Computing, 2023) solved the same problem on geode-fem's own meshes, written out with their physical tags, with the same materials, PEC surfaces and uniform 50 Ω lumped port. Its absorber is different: Palace has no UPML here, so the UPML shell is air and the outer wall carries a first-order absorbing boundary. On L1, Palace and geode-fem agree on the dip to 1.1 MHz at

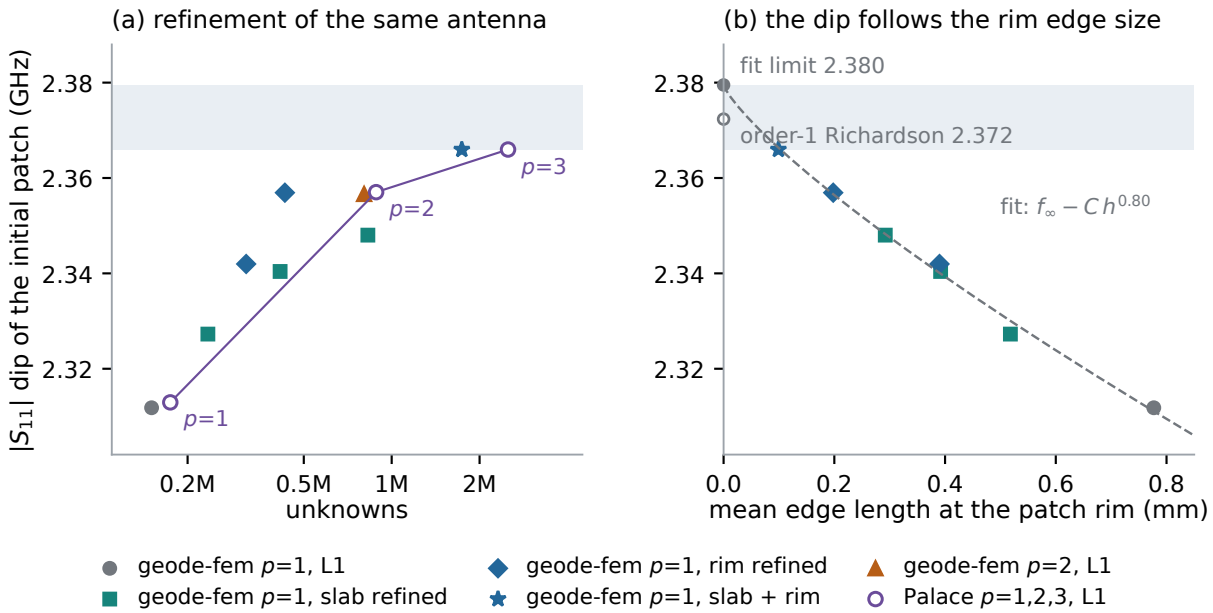


Figure 2: The initial patch's $|S_{11}|$ dip under refinement. (a) Against the number of unknowns: geode-fem at $p = 1$ on L1 and its slab, rim and combined refinements, geode-fem at $p = 2$ on L1, and Palace at $p = 1, 2, 3$ on the same L1 mesh. (b) Against the mean edge length at the patch rim, with the fit $f_\infty - Ch^{0.80}$ (dashed), its limit and the order-1 Richardson step on the 0.39/0.20 mm rim pair. The shaded band, 2.366–2.380 GHz, runs from the finest computed value to the fit's limit; it is an estimate, not a bound. Data: `v6/v6_summary.toml [x1]`.

$p = 1$ (2.3130 against 2.3118 GHz) and 0.4 MHz at $p = 2$ (2.3570 against 2.3566 GHz), and on the strip-fed design of Section 6 the P2/L1 −10 dB band reproduces to about 1 MHz. Palace at $p = 3$ on L1 gives 2.3660 GHz, 8.9 MHz above its $p = 2$ value and 0.02 MHz from geode-fem's finest rung (slab 0.30 plus rim 0.10 mm), 2.3659 GHz (Figure 2a). Two independent refinements of the same model, order in Palace and rim size in geode-fem, meet at the same number.

The mesh-limit resonance. Every refinement computed so far raises the dip. If that continues, the finest computed value, 2.366 GHz, is a lower estimate of the mesh-limit dip; this is an inference from the trend, not a proof. The rim fit puts the limit at 2.380 GHz and the order-1 Richardson step on the rim pair at 2.372 GHz. We therefore estimate the initial patch's mesh-limit dip at 2.366–2.380 GHz, 0 to 14 MHz above the finest computed models. The Hammerstad transmission-line estimate of the unloaded patch's resonance is 2.433 GHz (Hammerstad, 1975), 54 to 67 MHz above the estimated mesh-limit dip. The two quantities differ: the closed form gives the unloaded cavity resonance and ignores the probe and the finite ground, and we expect, but did not test, that the probe's reactance pulls the matched dip below that resonance. A port-free eigenfrequency would separate them and was not computed.

Absorber and air margin. The UPML conductivity $\sigma_0 = 1$ was chosen from a ladder of σ_0 and shell thickness, over which $|S_{11}|$ changes by at most 0.021 in magnitude. Re-meshing the strip-fed design of Section 6 with a 60 or 80 mm air margin instead of 40 mm moves its $p = 1$ dip by at most 2.4 MHz and its −10 dB band by at most 1.4 MHz, not monotonically. The Palace runs, with a different absorber, agree with geode-fem to the levels above. The absorber does not load the

Table 3: Solves to fixed worst-of-band levels on the L1 optimization model ($p = 1$), from $\mathbf{c} = 0$. Three CMA-ES entries are seeds (the first ran 114 evaluations, the others 190). "Best" is the best level in the run, "in 125" the best within the adjoint's 125 solves. Wall-clock to −9.5 dB on an r6i.4xlarge: measured for Palace, estimated for geode-fem. A dash is a censored entry: the run did not reach the level within its budget of 535, 785 and 690 solves (geode-fem CMA-ES seeds), 860, 780 and 660 (Palace CMA-ES seeds) or 945 (NEWUOA).

Optimizer	Forward	Solves to			Best	In 125	Wall to
		−9 dB	−9.5 dB	−10 dB	(dB)	(dB)	−9.5 dB (min)
L-BFGS, adjoint	geode-fem	30	35	60	−10.04	−10.04	11.1 (est.)
sep-CMA-ES	geode-fem	405	520	–	−9.56	−2.66	125 (est.)
		385	385	695	−10.02	−2.40	93 (est.)
		395	395	–	−9.77	−2.82	95 (est.)
sep-CMA-ES	Palace	405	805	–	−9.58	−2.66	172
		385	385	–	−9.89	−2.40	84
		395	570	–	−9.90	−2.83	124
NEWUOA (PRIMA)	Palace	415	420	–	−9.95	−1.52	90

antenna at the level of the refinement shifts.

Curvature. The same copper on a flat board (bend radius 10^6 mm, the same mesh node for node) resonates 0.3 MHz from the bent one at $p = 1$, and a complex frequency-shift fit at P2/L1 finds 0.28 MHz. The bend does not detune the patch. It changes its input impedance: on the strip-fed design the bent board has a 93.3 MHz $p = 1$ −10 dB band against 82.8 MHz flat, and a worst in-band $|S_{11}|$ at P2/L1 of −10.78 against −9.93 dB. The design was optimized on the bent board; a design optimized flat might recover the difference, and was not run.

5 The Gradient Against Gradient-Free Search

Setup. All optimizers solve one problem: the 38 parameters, the objective (3) on the unshifted band, the 0.25 volume guard and the L1 fixture at $p = 1$, starting from $\mathbf{c} = 0$ (−1.47 dB). The adjoint runs L-BFGS. Separable CMA-ES runs with population $\lambda = 14$ and initial step $\sigma = 0.5$ for 190 evaluations (one seed for 114), with three seeds, and with geode-fem or Palace as the forward solver. NEWUOA runs with Palace, initial trust radius 0.5 (equal to σ), final radius 10^{-3}, PRIMA's default $2n + 1 = 77$ interpolation points and 190 evaluations. For Palace, geode-fem applies the morph and the guard, writes the moved mesh with its tags, and Palace solves the five band frequencies directly at $p = 1$ with GMRES on 16 MPI ranks. NEWUOA is unconstrained, so a design that fails the guard receives a penalty above any feasible objective without a solve; this happened once in 190 evaluations.

Counting. One solve is one band frequency with one matrix factorization. The adjoint reuses the forward factorization for its adjoint solve and counts one solve per frequency; its start, which yields its first gradient, is counted. The gradient-free runs' separate evaluation at $\mathbf{c} = 0$ is not counted, except as NEWUOA's own first evaluation. Guard-rejected CMA-ES samples cost no solve. Each level is the worst of the five band points in that run's own forward model. Wall-clock is on one host type, an r6i.4xlarge (16 vCPU): measured for the Palace runs, which include mesh writes and Palace start-up, and estimated for the geode-fem runs as solves times the per-solve cost measured on that host, 14.4 s forward and 19.0 s forward plus adjoint. A Palace band evaluation costs 64.4 s against geode-fem's 72.1 s, so Palace's forward solve is about 11% faster.

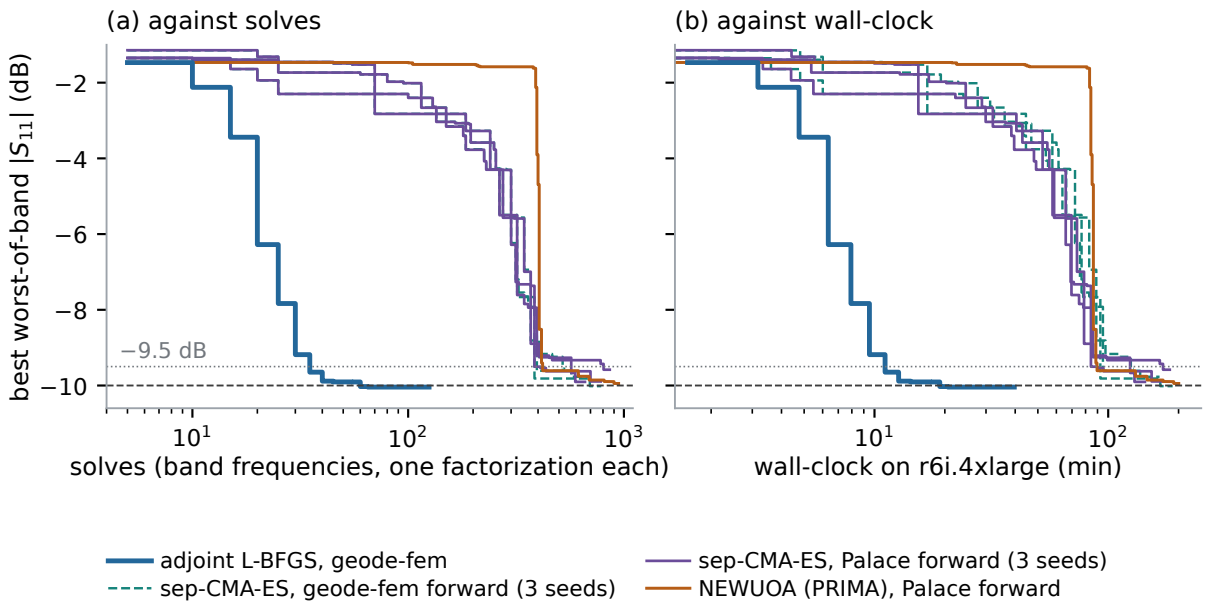


Figure 3: Best worst-of-band $|S_{11}|$ so far on the L1 optimization model, for the adjoint L-BFGS and the gradient-free runs of Table 3, (a) against solves and (b) against wall-clock on one r6i.4xlarge. Palace wall-clock is measured; geode-fem wall-clock is estimated from the measured per-solve cost on that host. Data: `copper/L1_copper_38.toml`, `v6/x4/`, `v6/palace_loop/`.

Results. Table 3 and Figure 3 give the comparison. To reach −9.5 dB the adjoint uses 35 solves; CMA-ES uses 385–520 with geode-fem and 385–805 with Palace, and NEWUOA 420, that is 11 to 23 times as many. In wall-clock the Palace-driven runs take 84–172 min (CMA-ES) and 90 min (NEWUOA) against an estimated 11.1 min for the adjoint, 7.6 to 15.5 times as long. The wall-clock ratio is smaller than the solve ratio because an adjoint evaluation costs 1.32 forward evaluations and Palace's forward solve is the faster one. Within the adjoint's whole budget of 125 solves, the gradient-free runs are between −1.52 and −2.83 dB. NEWUOA spends its first 77 evaluations (385 solves) building its interpolation model, which with $2n + 1$ points is close to one finite-difference gradient, and then steps to −9.61 dB within 8 evaluations; a smaller interpolation set was not tried.

The adjoint reaches −10 dB in 60 solves. One geode-fem CMA-ES seed of three reaches it, at 695 (11.6×), and then matches the adjoint's final value (−10.02 against −10.04 dB). The other two seeds are censored: they miss within their budgets of 535 and 690 solves, which supports only ratios of at least 8.9 and 11.5. Every Palace-driven run is censored as well, missing within 190 evaluations (660 to 945 solves), and under Palace's own model the adjoint's final design sits at −9.80 dB, also short of −10 dB. We therefore state the speedup at −9.5 dB and derive none from the −10 dB column. That column needs care for a second reason. The two forward models agree to 0.004 dB on the 41 designs both evaluated (the first CMA-ES generation of each seed, 42 samples with one rejected by the guard in both), but the difference grows at optimized designs, where the band edge sits on a steep slope: Palace reads the adjoint's final design at −9.80 dB against geode-fem's −10.04 dB, and geode-fem reads the four Palace-optimized designs at −9.81 to −9.89 dB, within 0.25 dB of Palace's own values. A 0.2 dB model difference is the size of the gap between reaching and missing −10 dB, so that column compares two slightly different targets. The −9 and −9.5 dB columns do not depend on it.

What the gradient buys, and what it does not. This is an ablation of the gradient: the geode-fem CMA-ES runs share the adjoint's forward model, objective, basis and guard, and the Palace runs show that a faster forward solver does not close the gap. On this problem the gradient buys speed, about an order of magnitude in solves, and not a better optimum. Each gradient-free optimizer was run at one setting ($\sigma = 0.5$, $\lambda = 14$; PRIMA's defaults), so a tuned gradient-free method would do better than these rows. All levels are on the L1 model at $p = 1$, which sits about 45–57 MHz below the refined models (Section 4); none is a level of the refined model.

A four-parameter rectangle. The 38-parameter basis is not needed to meet the target. Collapsing it to the patch width and length plus the two feed parameters, with the same objective, guard and band shift as the strip-fed design's final step and the same feed re-basing (Section 6), the adjoint widens the patch from 38 to 46.1 mm and reaches a worst in-band $|S_{11}|$ of -10.33 dB at P2/L1, against -10.78 dB for the 38-parameter design on the same model. Its $p = 1$ -10 dB band is 86.9 MHz against 93.3 MHz, so the free outline adds about 6 MHz of bandwidth. The rectangle started with the 38-parameter chain's final band shift instead of finding it, so its 340 driven solves are not comparable with the chain's 655, which include the steps that found the shift.

6 Design Procedure

Frequency space mapping. A design optimized on L1 resonates higher on a finer model. We model the fine response as the coarse response shifted in frequency, $S_{\text{fine}}(f) \approx S_{\text{coarse}}(f - \delta)$, fit δ by least squares over the fine model's points, and re-optimize on the coarse model for the band shifted down by δ. For the cold L1 optimum of Section 5 the fitted L1-to-P2/L1 shift is 47.85 MHz, and the root-mean-square complex misfit falls from 0.23–0.38 unshifted to 0.018–0.025 shifted over the designs fitted. That shift is about 3 MHz from the initial patch's own L1-to-P2/L1 shift (44.8 MHz), so the copper motion adds little detune of its own. Without the correction, the cold L1 optimum (-10.04 dB on L1) gives -4.76 dB at P2/L1. This ablation is confounded, because the corrected chain also re-bases the feed and widens the patch; a cleaner comparison is a design optimized on the finer mesh L2 with a 30 MHz shift and no re-base, which gives -9.92 dB at P2/L1, against a fitted L2 shift of 29.33 MHz.

Feed re-basing. Moving the feed toward the $+y$ edge compresses the mesh between them, and the cold runs stop there at the volume guard. We then regenerate the mesh with the port at its moved position and carry the design over by shifting the two feed coefficients. After re-basing, the optimizer widens the patch until the guard binds in the air 5.5 mm outside a $\pm s$ edge, which is a non-radiating edge. The widening, not the feed, is where every later descent stops.

Three designs. Table 4 lists the designs. Design A is the end of the space-mapping chain centred for P2/L1 with the feed width free, which widened the port into a 4.27 mm strip. At P2/L1 it reaches -10.78 dB with a 94.9 MHz band, 2.3938–2.4887 GHz. At Palace $p = 3$ the band moves up to 2.4029–2.4975 GHz, the lower edge misses 2.400 GHz by 2.9 MHz, and the worst in-band value is -9.56 dB: a design centred on P2/L1 misses once the rim is resolved. Design B re-runs A's last step with the probe width pinned at 1.6 mm and reaches -10.62 dB at P2/L1 (-10.65 dB with a 1.27 mm probe), so the wide strip was not what carried the bandwidth. Design C re-centres B for the refined model with a band shift of 57.0 MHz: 44.94 MHz from L1 to P2/L1, 8.7 MHz for design A's $p = 2$ to $p = 3$ dip shift in Palace, and 3.4 MHz further down because every refinement so far has moved the band up. A parallel run at 53.6 MHz was set aside: adding the initial patch's step of about 9 MHz from the 0.2 mm-rim mesh to Palace $p = 3$ to each design's lower edge on that mesh predicts a

Table 4: The designs. δ is the band shift of the final descent. "Solves" counts the driven solves of that descent as recorded, including its gradient checks. "Fine model" is the model the design was centred for.

Design	Feed	δ (MHz)	Steps	Solves	Fine model
A	4.27 mm strip (free width)	44.94	9	125	P2/L1
B	1.6 mm probe (pinned)	44.94	10	130	P2/L1
C (final)	1.6 mm probe (pinned)	57.0	10	125	refined rim, Palace $p = 3$

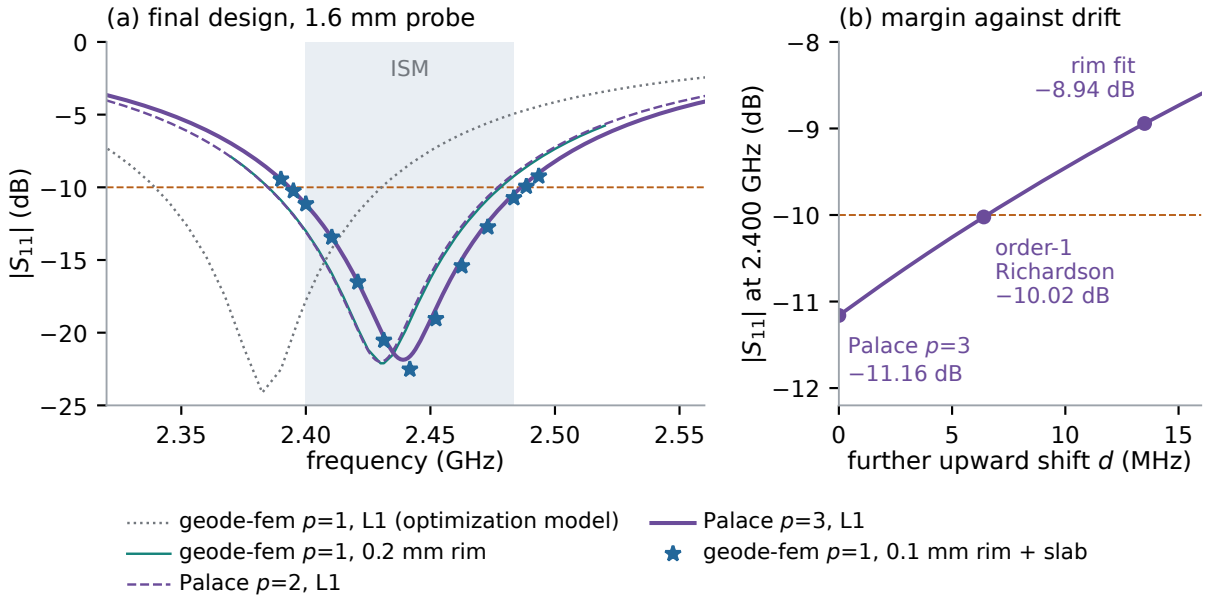


Figure 4: (a) Return loss of design C. Palace at $p = 3$ and $p = 2$ on the design's L1 mesh (1 MHz grids); geode-fem at $p = 1$ on the combined 0.1 mm-rim and 0.3 mm-slab mesh (13 points) and on the 0.2 mm-rim mesh; and geode-fem at $p = 1$ on L1, the optimization model, on which the design sits 57 MHz low by construction. The dashed line is −10 dB and the shaded band is the ISM band. (b) $|S_{11}|$ at 2.400 GHz if the whole Palace $p = 3$ response moves up by a further d, read from the measured response at $2.400 - d$ GHz, with the two drift estimates of Section 4 marked. Data: `recentre/`.

lower-edge margin at $p = 3$ of about 2.6 MHz for it and about 6 MHz for C. The descent to C ran 10 steps and 125 solves, from −8.73 to −10.64 dB on its shifted L1 band, and stopped on the guard with the gradient norm down from 0.143 to 2.2×10^{-3}; the gradient matched finite differences to 1.2×10^{-6}. As built, C is 29.00 mm long and 49.72 mm wide on the centre line, with a barrel outline whose $\pm s$ edges bow out most at mid-length, and its 1.6 mm probe sits 5.14 mm from the $+y$ edge.

7 Results

On the finest models. On Palace at $p = 3$ design C has a worst in-band $|S_{11}|$ of −10.46 dB, at 2.4835 GHz, and a contiguous −10 dB band from 2.3935 to 2.4864 GHz, 93.0 MHz wide, which covers the whole 83.5 MHz ISM band with 6.5 MHz to spare at 2.400 GHz and 2.9 MHz at 2.4835 GHz (Table 5, Figure 4a). At 2.400 GHz it is at −11.16 dB. geode-fem's finest mesh, the combined 0.1 mm rim and 0.3 mm slab (1.74M unknowns), gives −10.73 dB and a 94.7 MHz band, 2.3934–2.4881 GHz. The two lower edges agree to 0.02 MHz, as the two codes agreed on the initial patch's dip. The upper edges differ by 1.7 MHz, partly because geode-fem's upper crossing is interpolated across a 5 MHz gap and Palace's across 1 MHz. The two models are not equally independent of the design. Palace

Table 5: Design C on each model, in order of refinement, with design A on Palace $p = 3$ for comparison. "Worst" is over the nine in-band points (2.400 to 2.4835 GHz in 10.4375 MHz steps). Band crossings are interpolated linearly in dB. Margins are 2.400 GHz minus the lower crossing and the upper crossing minus 2.4835 GHz. Model-dependent values.

Model	DOFs	Worst (dB)	−10 dB band (GHz)	Coverage	Margins (MHz)
geode-fem $p = 1$, L1	150,541	−4.94	2.3393–2.4304	36%	60.7 / −53.1
geode-fem $p = 2$, L1	804,336	−9.08	not swept	not swept	not swept
Palace $p = 2$, L1	884,536	−9.08	2.3847–2.4770	92.2%	15.3 / −6.5
geode-fem $p = 1$, rim 0.2	431,528	−9.23	2.3849–2.4782	93.6%	15.1 / −5.3
geode-fem $p = 1$, rim 0.1 + slab	1,738,279	−10.73	2.3934–2.4881	100%	6.6 / 4.6
Palace $p = 3$, L1	2,503,077	−10.46	2.3935–2.4864	100%	6.5 / 2.9
Design A, Palace $p = 3$	2,729,331	−9.56	2.4029–2.4975	96.5%	−2.9 / 14.0

$p = 3$ informed the band shift, through design A's $p = 2$ to $p = 3$ step; geode-fem's combined mesh was not used in choosing the design. On the coarser models the design sits high by construction and misses at the upper edge: at P2/L1 Palace and geode-fem agree to 0.003 dB on its worst in-band value (−9.076 and −9.079 dB). From $p = 2$ to $p = 3$ the design's dip rises 9.1 MHz and its lower band edge 8.7 MHz; the initial patch's dip rises 8.9 MHz over the same step, so the design shifts like the patch.

The refined geode-fem meshes were made with a rim refinement laid around the initial 38×29 mm outline; measured on the moved meshes, the mean rim edge length around the moved outline is 0.21 mm and 0.10 mm, unchanged by the design. On the combined mesh the default air morph inverts elements 0.1 mm above the patch edge, so the air there was moved on cylinders instead, which keeps the copper, slab and ground identical; on the 0.2 mm-rim mesh the two air motions agree to 0.09 MHz at the lower edge and 0.015 dB in the worst in-band value.

Margin against the remaining drift. If the whole band moves up by a further d, design C keeps the ISM band covered at the lower edge while $d \leq 6.5$ MHz; the upper edge only gains. The estimates of the remaining drift above the finest models are those of the initial patch (Section 4). For the order-1 Richardson estimate, $d = 6.4$ MHz, the lower edge reaches about 2.3999 GHz and 2.400 GHz sits at −10.0 dB, just inside. For the rim-fit estimate, $d = 13.5$ MHz, the lower edge reaches about 2.4070 GHz and 2.400 GHz sits at −8.94 dB, a miss (Figure 4b). These two values are the measured $p = 3$ response shifted, not computed. The order-1 estimate may be low. The only data point that tests it is the combined rung of Table 2, which moved the initial patch's dip 9.0 MHz where the order-1 step on the rim pair predicts 7.7 MHz; that rung also refines the slab, so the comparison is confounded. So the design meets the target with margin on the most accurate computed models, stays inside the band for the lower part of the drift estimate, and misses at 2.400 GHz if the drift reaches the rim-fit value. Covering the whole 0 to 14 MHz range would need about 97.5 MHz of −10 dB bandwidth, 4.5 MHz more than design C has.

The guard limits the bandwidth. The descents that produced designs A, B and C each stop on the 0.25 volume guard in the air outside the widening patch, not at a stationary point of the objective, so the 91–95 MHz bandwidth and with it the slack are what the guard allows. Regenerating the mesh at the widened outline would relieve the guard; it was not run.

Radiation efficiency and gain. For the strip-fed design A, whose outline matches C's to 0.2 mm in length and 0.03 mm in width, we computed the accepted port power, the Poynting flux through a closed surface in the air box, the FR-4 loss and the broadside gain by a near-to-far-field transform, at $p = 1$ on L1 and on the finer L3 mesh. The power balance closes to 2.3% on L1 and 1.2% on L3. The radiation efficiency is 0.61–0.62 on L1 and 0.62 on L3 across the band, the dielectric loss fraction 0.36, and the broadside gain 4.4–4.6 dBi with a front-to-back ratio of 12–15 dB. With PEC conductors this efficiency counts radiation against dielectric loss only; copper loss would lower it. These values were not recomputed for design C, whose feed differs.

Bandwidth against a closed form. On the voltage-standing-wave-ratio-2 criterion, Palace gives design A a bandwidth of 100.3 MHz at $p = 2$ and 101.0 MHz at $p = 3$ (design C: 99.0 MHz at $p = 3$). The closed-form formulas of Jackson and Alexopoulos (1991), in the form of Jackson's course notes (Jackson, n.d.), give 70.4 MHz for a 49.75×28.80 mm rectangle on this substrate, and 67.9 MHz at the area-equivalent width of 45.4 mm. The gap is in the radiation Q. The only loss in the model is $\tan\delta$, so the FEM's dielectric loss fraction bounds its total Q from below, $Q_t \geq 0.362/\tan\delta = 18.1$, and with $\eta = 0.615$ its radiation Q is at least 29; the closed form gives $Q_t = 24.5$ and a radiation Q of 48.0, with radiated and dielectric fractions of 0.51 and 0.49 against the FEM's 0.62 and 0.36. The model patch radiates more strongly than the infinite-ground closed form; the finite 62×53 mm ground is a plausible cause that we did not test.

Cost. The final descent took 26 min on a 32-vCPU cloud host shared with a second run. The Palace $p = 3$ sweep of design C (281 frequencies from an adaptive reduced-order model built on 12 full solves at 2.5M unknowns) took 3.5 h on 16 MPI ranks. On the combined mesh geode-fem's direct solver took about 8 min per frequency at 204 GiB peak memory, which is why that mesh served for verification only.

8 Limitations and Future Work

Feed and losses. The probe is a uniform lumped port of the pin's width across the substrate. A coaxial feed with its own reactance is not modelled, conductors are PEC, and the design has not been fabricated or measured.

Refinement. The remaining drift is estimated, not computed, and design C covers only the lower part of its range. The adjoint is first order; a $p = 2$ shape adjoint, or optimizing on a rim-refined mesh directly, would remove the coarse-to-fine shift that space mapping corrects here. A port-free eigenfrequency, which would separate the feed reactance from the patch resonance, was not computed, and Palace ran on L1 only.

Scope of the comparison. The optimizer comparison is one problem, one starting point, one setting per gradient-free optimizer and one forward model level. A smaller NEWUOA interpolation set, Bayesian optimization or a tuned CMA-ES might narrow the gap. The comparison measures search cost on the coarse model; it does not test whether the gradient helps at the refined level.

Comparison with FDTD. A matched-geometry Meep (Oskooi et al., 2010) model of the initial antenna, run at three grid resolutions down to 0.25 mm cells, disagrees with geode-fem and Palace. In the sampled 2.2–2.7 GHz window its only deep $|S_{11}|$ feature sits at the 2.20 GHz lower edge, which suggests a resonance at or below 2.20 GHz, while geode-fem and Palace on their refined models (second order or rim-refined) place the dip at or above 2.357 GHz (2.31 GHz on L1 at first order);

and under refinement the two move in opposite directions, the finite-element dip upward and Meep's minimum downward. The cause has not been identified and is being diagnosed. Until it is, we make no comparison of cost or accuracy with FDTD, including a conformal FDTD scheme (Dey and Mittra, 1997; Zagorodnov et al., 2003).

9 How geode-fem Was Designed and Built

geode-fem implements the whiteroom operator specification of Crutcher Dunnavant (Dunnavant, 2026), whose four layers restate Palace's cited source as pure functions, fused primitives, tensor operations and a small graph-evaluation calculus. He also gave design input on the solver's operator layer and contributed code: the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate. The rest of the code, the experiments and the text of this paper were produced by AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by Squad, a message room the agents share. Nearly every change was reviewed by a separate agent session. The agent reviews of the experiments behind this paper recomputed the reported numbers from the archived records, and the script that checks this paper's text against those records was mutation-tested. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

10 Reproducibility

Every result in this paper is read from an archived, scripted artifact.[1] Table 6 lists them, relative to `benchmarks/patch_antenna_conformal/`. Each run record holds the exact command and the SHA-256 of its mesh; the meshes are regenerated from YAML recipes in `reference/gmsh/` with gmsh 4.15.2, and runs on re-based fixtures need the `--feed-support-legacy` flag. Most records carry no generating commit and one carries an intermediate one; the code that produced them is merged at geode-fem commit `99646f06`. Palace is commit `fba6a5b`, built from `reference/palace/docker/Dockerfile`. PRIMA was built from source at libprima commit `43863c69` (Python package `0.7.3.dev1801+g43863c69c`); the compiler was not recorded. The geode-fem solves ran on AWS Graviton4 hosts (r8g.4xlarge and r8g.8xlarge), and the Palace solves and the cost measurements on r6i.4xlarge. The figure scripts in `figures/src/` read the artifacts directly, `provenance.md` in this paper's directory maps every number to its source field, and `verify_evidence.py` checks the text against them.

11 Conclusion

On a 38-parameter copper design of a bent 2.4 GHz patch, an adjoint gradient reaches a worst-of-band $|S_{11}|$ of -9.5 dB on the coarse optimization model with an order of magnitude fewer solves than two gradient-free optimizers, whether they are driven by the same solver or by a faster independent one; at -10 dB most gradient-free runs are censored at their budgets. The gradient buys speed, not a better optimum, and a four-parameter rectangle, given the final band shift in advance, would also have met the target on the second-order model. Getting the target to hold on a fine model needed the forward model resolved at the patch rim, where an independent solver agrees with geode-fem on the resonance, and one more space-mapping step. The resulting design, with a 1.6 mm probe on a standard board, covers the ISM band at -10 dB on both fine models with 6.5 MHz of margin at the

[1] The geode-fem source is public at `https://github.com/rjwalters/geode-fem`. Archival DOI to be added at deposit.

Table 6: Artifacts behind each result, relative to `benchmarks/patch_antenna_conformal/`.

Result	Artifact
Fixture, coarse ladder, absorber ladder	`baseline_2g4_L1.toml`, `convergence.toml`, `upml_ladder_L1.toml`
Rim and slab refinement, Palace cross-check, curvature, radiation, air margin (Section 4)	`v6/v6_summary.toml` from `scripts/v6_summary.py`; `v6/x1/`, `v6/x2/`, `v6/x3/`
Gradient-free baselines (Section 5)	`v6/palace_loop/palace_loop_summary.toml` from `scripts/palace_loop_summary.py`; `v6/x4/`, `v6/palace_loop/`; `copper/L1_copper_38.toml`
Space mapping, designs A and B (Section 6)	`copper/`, `copper/g2_summary.toml`, `scripts/copper_shift_fit.py`; `v6/x5/`
Design C and its checks (Section 7)	`recentre/recentre_summary.toml` from `scripts/recentre_summary.py`; `recentre/`, `recentre/palace/`
FDTD comparison (Section 8)	`reference/meep/docker/AWS_R4.md`; `benchmarks/fdtd_density_baseline/meep_runs/aws_r4/resolution_trend.json`
Generator and tests	`crates/geode-core/examples/patch_conformal_nested_dof.rs`; `crates/geode-core/tests/patch_conformal_*.rs`

lower edge. That margin covers the order-1 estimate of the remaining drift, which may be low, but not the rim-fit estimate, and the volume guard, not the objective, limits it.

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